

The background of the image features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

AWS re:Invent

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Evaluating the cost of infrastructure changes

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AWS



Why are we here today?



IT/operations



Engineer/developer/cloud architecture



FinOps, CCoE

Traditional technology consumption



Model

Engineers as requesters

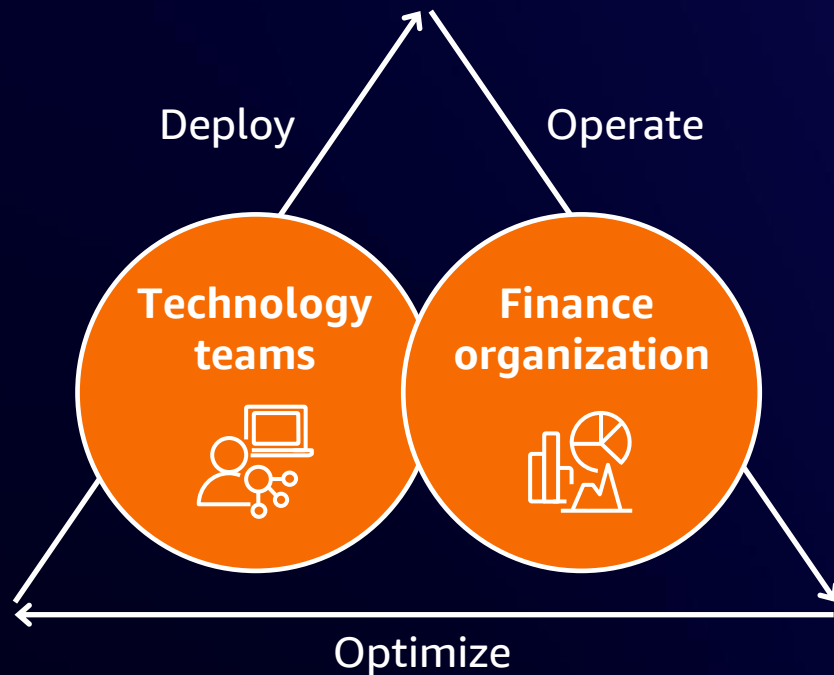
Procurement as approvers

Step function spend and capacity forecasting

Long procurement cycles

High cost of failure

Cloud technology consumption



Model

Engineers and finance team acting as one

Match capacity with demand

Procurement is instant

Low cost of failure

Adapting traditional processes to be cloud ready



Forecasting

Mapping to AWS

Dynamic, rolling forecast

Cross-functional alignment



Control

Consolidated billing

Governance and agility

Simplified procure-to-pay



Traceability

Clean account structure

Meaningful cost reporting

Ready for chargeback
and showback



Oversight

Cost-aware architecture

Match demand with supply

Various purchase options

Why cloud financial governance?

SAFEGUARDING YOU FROM ANYTHING THAT CAN GO WRONG FROM GOING WRONG



Eliminate shadow IT spend



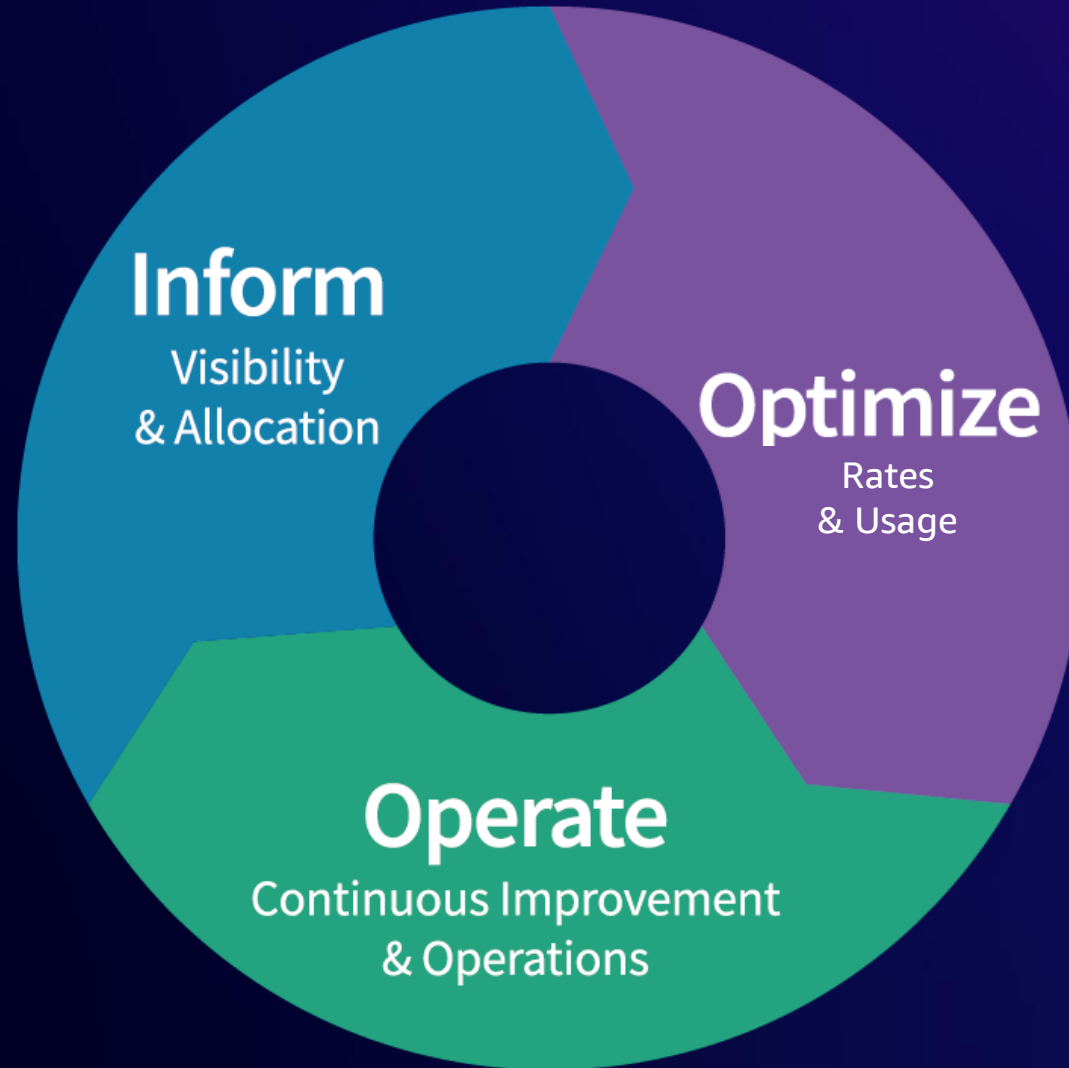
Standardize how cloud resources are procured and deployed



Establish fiscally responsible cloud usage at scale

FinOps

FinOps phases





Dr. Werner Vogels VP and CTO at Amazon.com

You build it, you run it.

This brings developers...into day-to-day contact with the customer. This customer feedback loop is essential for improving the quality of the service.

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What is GitOps?

GitOps is an approach where infrastructure is defined as code (IaC) alongside application code, versioned in a git repository, and then provisioned following the same merge request process as application software code

**Infrastructure
as code**

hosted just like
software code

+

Git

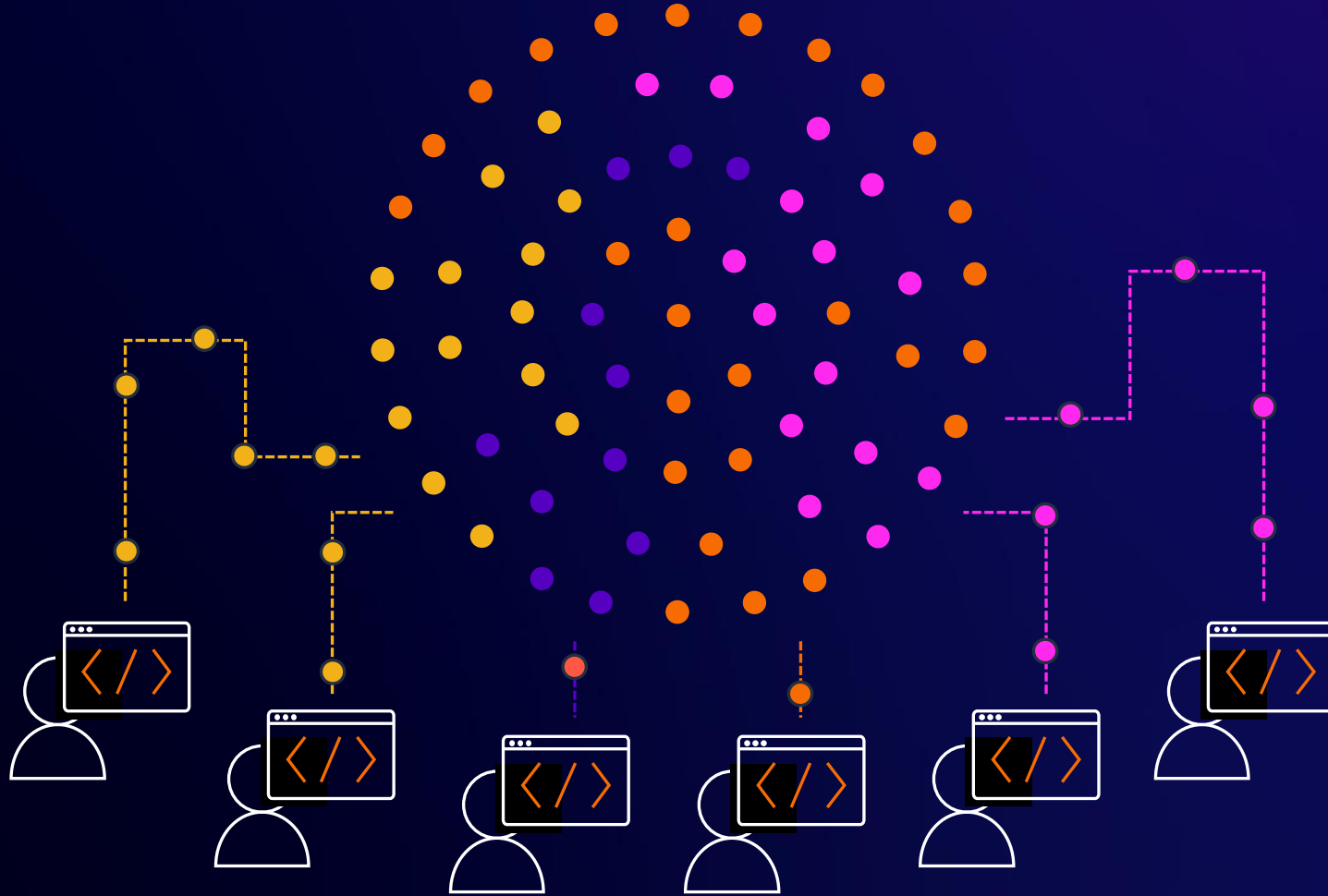
versioning and pull
requests workflows

+

CI/CD

for automated
build, test,
and deploy

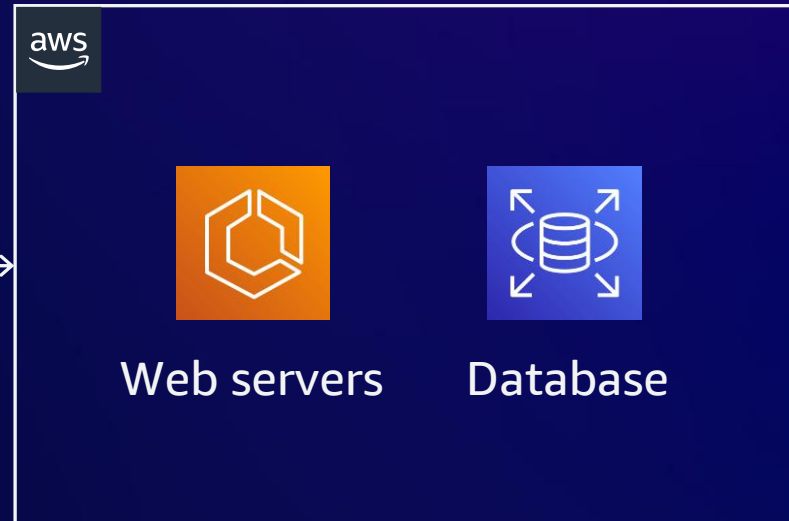
What is infrastructure as code?



Infrastructure as code

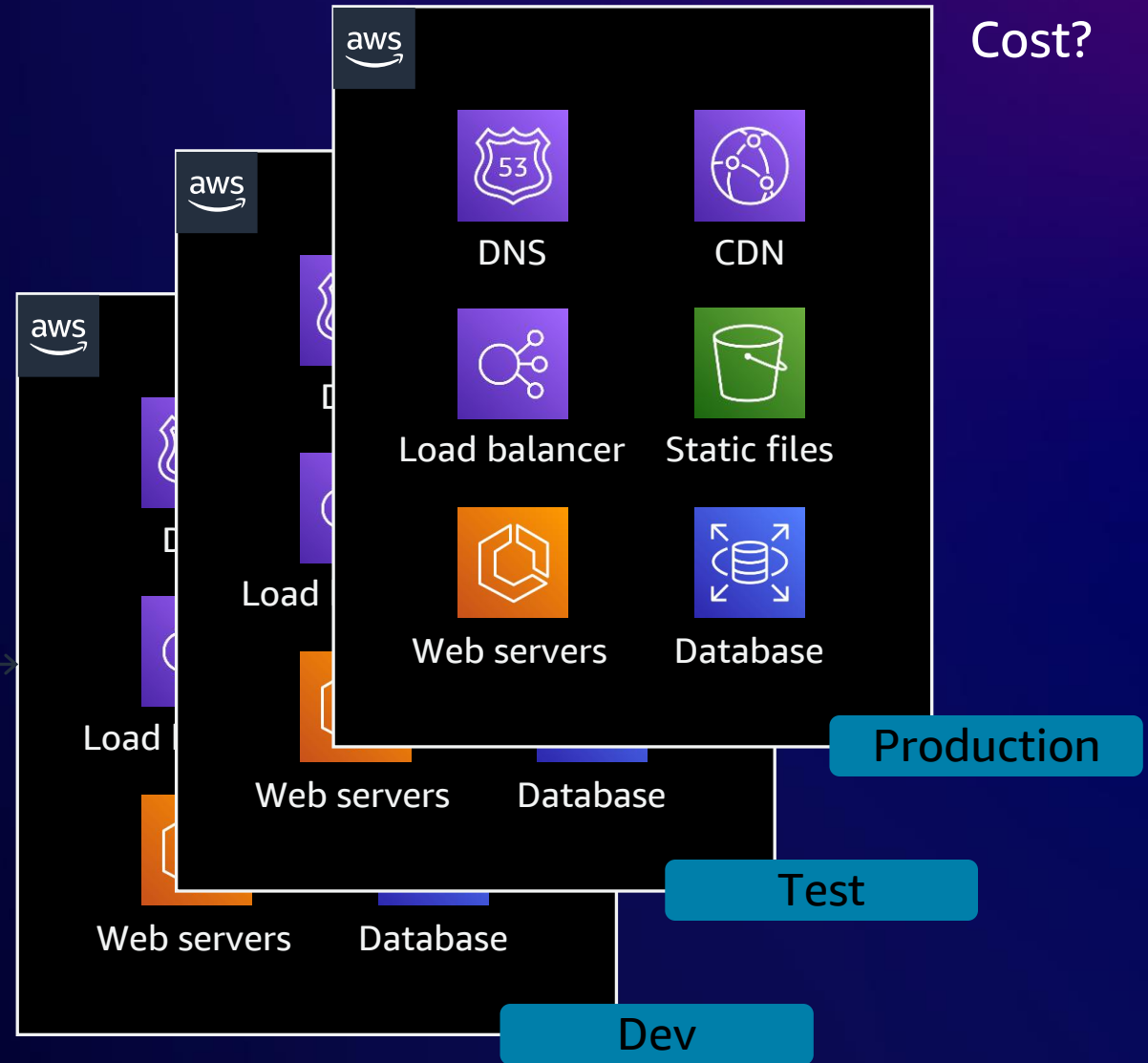
```
resource "aws_db_instance" "db" {  
  allocated_storage = 10  
  ...  
}  
  
resource "aws_ecs_service" "ecs" {  
  name = "ecsApp"  
  desired_count = 5  
  ...  
}
```

Cost – \$100



Environments as code

```
module "app" "prod": {  
  loadbalancer: {  
    server: {...}  
  },  
  cdn: {...},  
  database: {...},  
  dns: {...}  
  static: {...}  
}
```





IaC code → Cloud cost

You build it, you run it

(Dr. Werner Vogels)

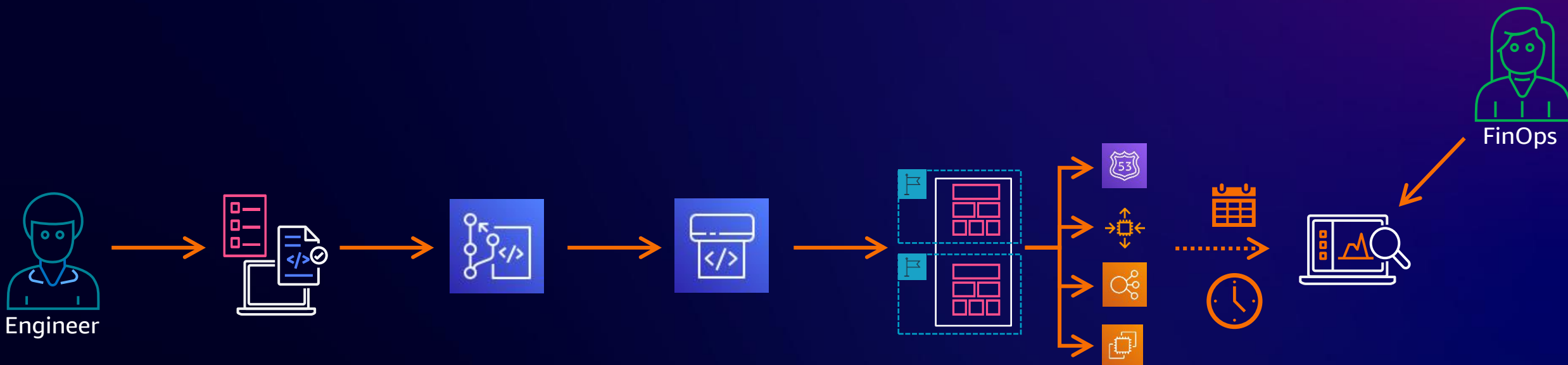
and you pay for it

(me)

GitFinOps



Classic GitOps



1

Code

2

Commit
git push/PR

3

Execute CI/CD
test and plan
deployment

4

Resources
provisioned
through IaC
tools

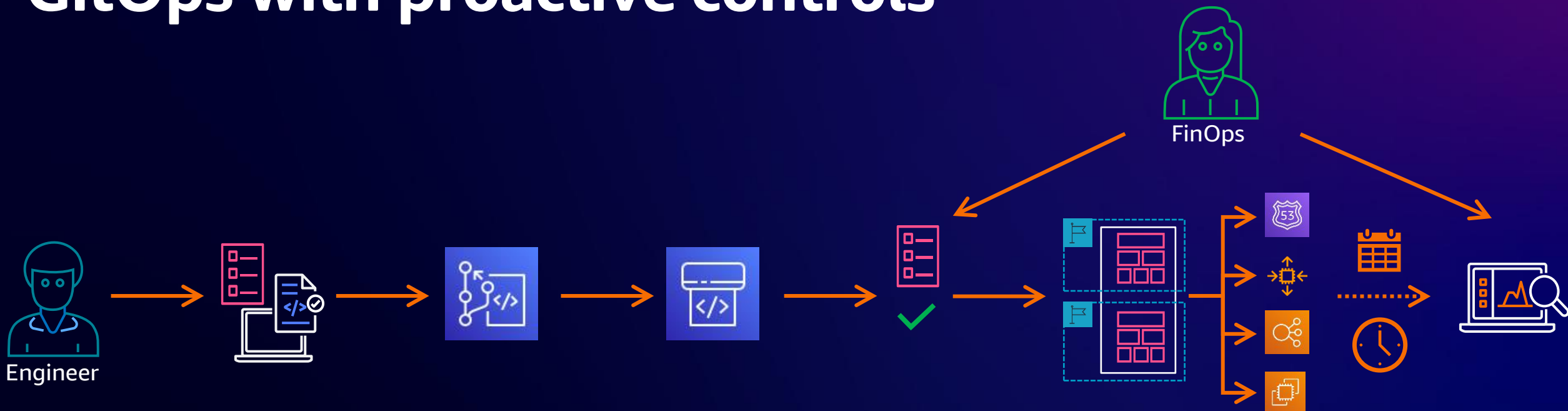
5

New costs are
reflected in AWS
Cost
Explorer/AWS
CUR

6

Costs allocated
to teams;
budgets
adjusted

GitOps with proactive controls



1

Code

2

Commit
git push/PR

3

Execute CI/CD
test and
deployment

4

Evaluate
approve costs
changes

5

Resources
provisioned
through IaC
tools

6

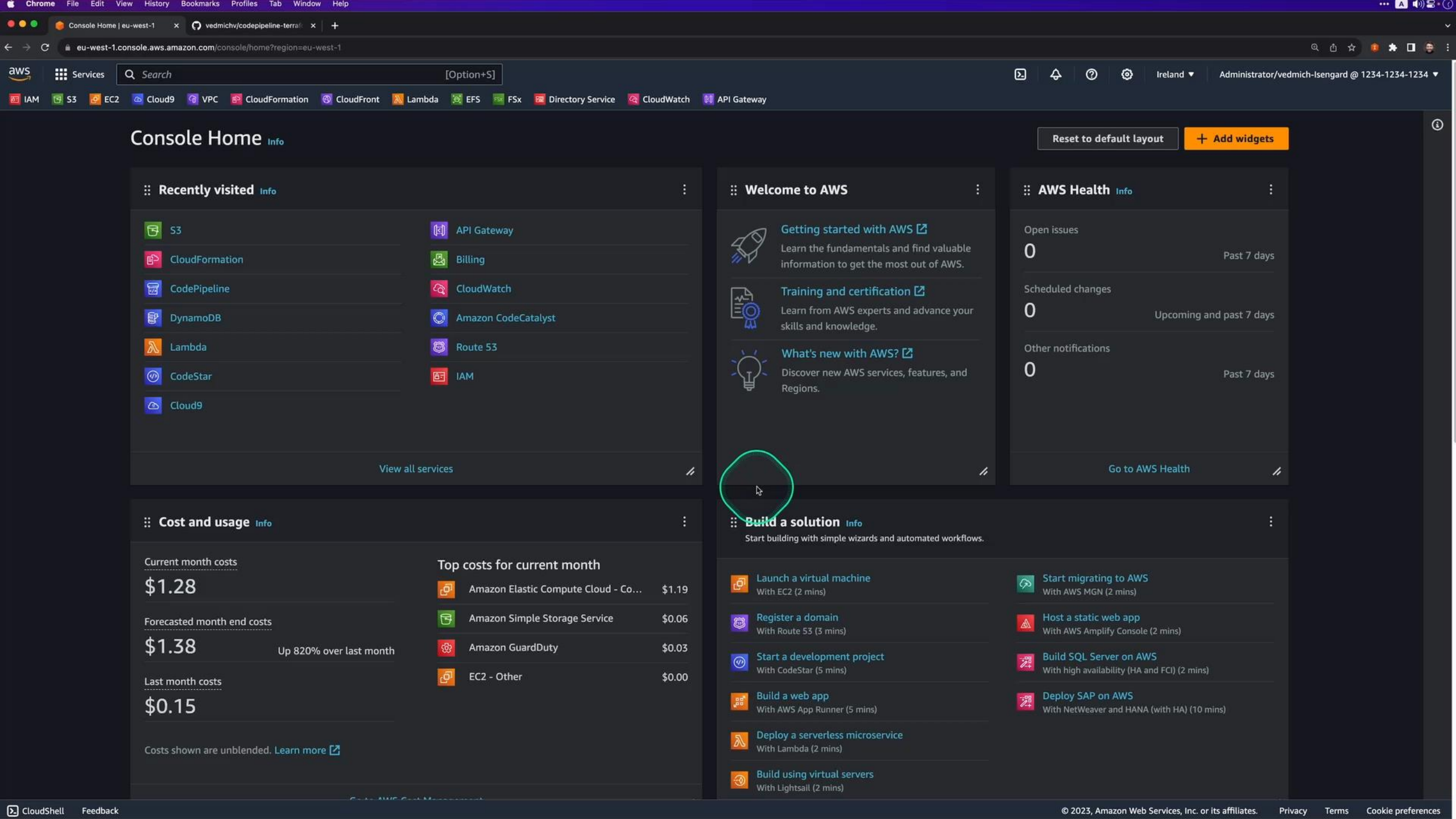
New costs are
reflected in AWS
Cost Explorer
/AWS CUR

7

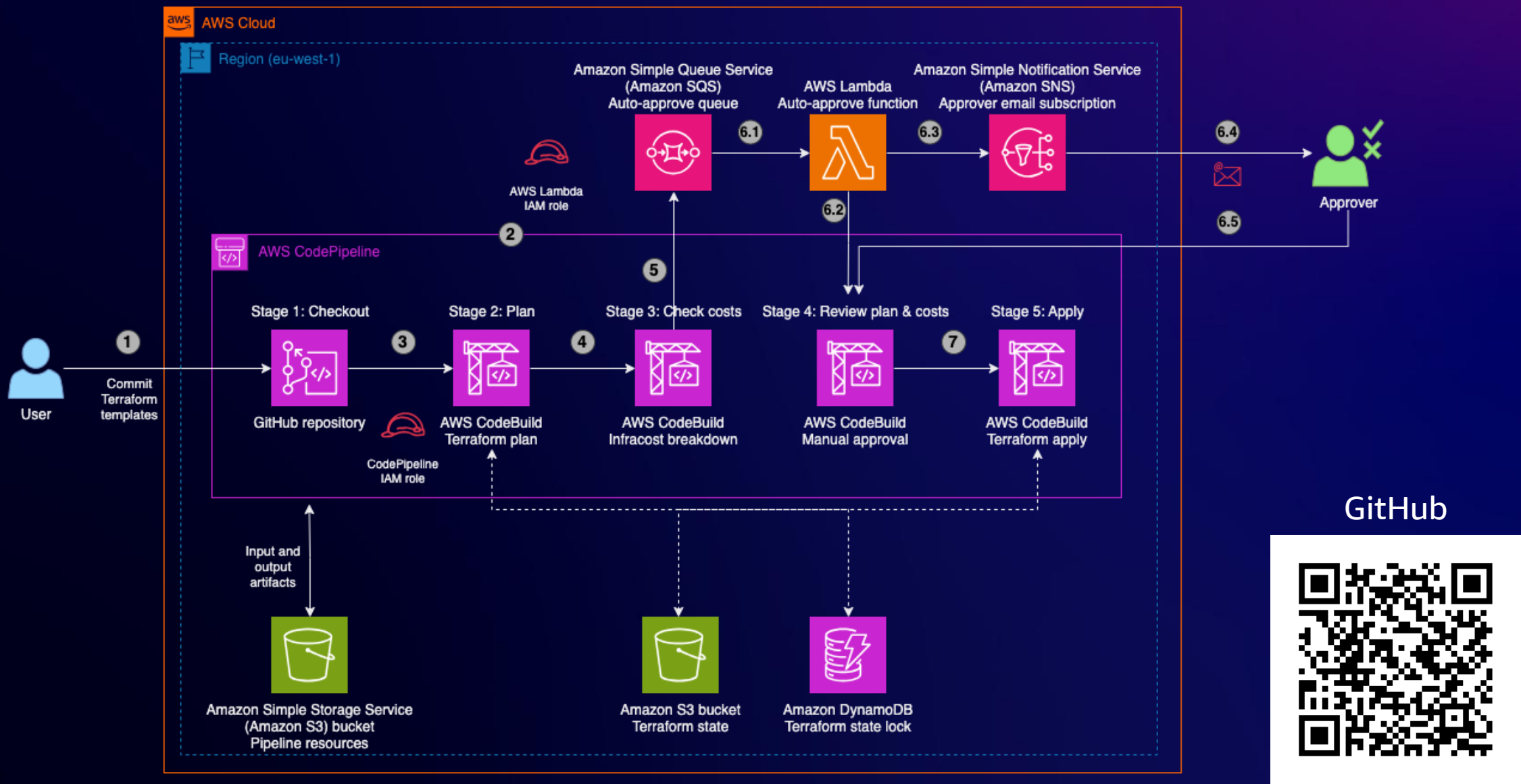
Costs allocated
to teams;
budgets
adjusted

Demo





Solution architecture



Thank you!



Please complete the session survey in the mobile app

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